
Java XML Application (Item example)

1. A shop contains many items. Each item has a **code**, **name** and **price**. You are required to write an XML code to store data in a file **items.xml**. Use **Shop** as root node and **Item**, **code**, **name** and **price** as tags.

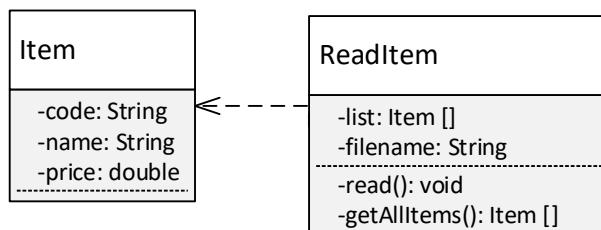
Code	Name	Price
001	Bread	12.99
002	Rama	40.99
003	Coke	16.49

```
<?xml version="1.0" encoding="UTF-8"?>
<!--A shop with many items; each item has code, name and price -->
<!DOCTYPE Shop SYSTEM "items.dtd">
<Shop>
  <Item>
    <code>00001</code>
    <name>Coke</name>
    <price>24.99</price>
  </Item>
  <Item>
    <code>00002</code>
    <name>Bread</name>
    <price>18.99</price>
  </Item>
  <Item>
    <code>00003</code>
    <name>Milk</name>
    <price>15.00</price>
  </Item>
</Shop>
```

2. The file **items.dtd** contains the code which defines the structure of the XML in items.xml. Write the code.

```
<!ELEMENT Shop (Item)*>
<!ELEMENT Item (code, name, price)>
<!ELEMENT code (#PCDATA)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT price (#PCDATA)>
```

3. A class **ReadItem** reads the values of each item in the XML file (**items.xml**) to create an instance of a class **Item** and store in a list. **ReadItem** and **Item** are shown below.



Item

- A POJO with instance variables **code**, **name** and **price**

ReadItem

list: an array list to store the instances of **Item** from item.xml

filename: the name of a file to open

read(): a method to open the file, read the XML, and store in list; it also closes all streams and handle exceptions

getAllItems(): a method to return the list of **Item** instances

4.1. Write the classes Item and ReadItem

Item.java

```
package item;

public class Item {
    private String code;
    private String name;
    private double price;

    public Item() {

    }

    public Item(String code, String name, double price) {
        this.code = code;
        this.name = name;
        this.price = price;
    }

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public String toString() {
        return "Code:" + code + "\tName:" + name + "\tPrice=R" + price;
    }
}
```

ReadItem.java

```
package itemwithif item;
import java.util.*; //for ArrayList
import java.io.*; // for FileReader
import javax.xml.stream.*; //XML
public class ReadItem {
    static ArrayList<Item> list=new ArrayList<Item>(); //to store Item object we
read
    String filename;
    public ReadItem(){

    }
    public ReadItem(String filenamePar){
        filename=filenamePar;
    }

    public void read()throws Exception{
        //1. Create stream;
        //2. Read data from XML file
        //3. Process the data
        //4. Close the stream
        //NOTE: Handle exceptions

        FileReader fr=new FileReader(filename); //connect to xml file
to read from
        XMLInputFactory fac=XMLInputFactory.newInstance();

        XMLStreamReader xmr=fac.createXMLStreamReader(fr);

        Item item=null;
        String value=null; //to store the value we read
        int event=0;
        while(xmr.hasNext()){ //while there an item
            event = xmr.next(); //get to the next
            //if event is the Item start element
            if((event==XMLStreamConstants.START_ELEMENT)&&
(xmr.getLocalName().equals("Item"))){
                item=new Item();
            }
            if(event==XMLStreamConstants.CHARACTERS) {
                value=xmr.getText();
            }

            if((event==XMLStreamConstants.END_ELEMENT)&&(xmr.getLocalName().equals("code"))){
                item.setCode(value);
            }

            if((event==XMLStreamConstants.END_ELEMENT)&&(xmr.getLocalName().equals("name"))){
                item.setName(value);
            }

            if((event==XMLStreamConstants.END_ELEMENT)&&(xmr.getLocalName().equals("price"))){
                item.setPrice(Double.parseDouble(value)); //why
the conversion
                //item.setPrice(Double.valueOf(value));
            }
        }
    }
}
```

```

        }

        if((event==XMLStreamConstants.END_ELEMENT)&&(xmr.getLocalName().equals("Item"))) {
            list.add(item);
        }
    }
    //close streams
    fr.close();
    xmr.close();

}

public ArrayList<Item> getAllItems() { //returns the list
    return list;
}

}

```

Tester.java

```

package item;
public class Tester{
    public static void main(String[] args) throws Exception {

        ReadItem ri=new ReadItem("items.xml");//
        ri.read();//
        for(Item i: ri.getAllItems()) { //get the list and for each item
            System.out.println(i); //print;
        }

    }

}

```

